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SUBJECT : DSA0604 - DATA HANDLING AND VISUALISATION

LAB EXPERIMENTS - 16 TO 20

Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

1 28 Female 50000

2 35 Male 60000

3 42 Female 75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

2. Generate a pie chart to represent the distribution of customers by gender.

3. Build a table to show the demographic information for each customer. Label the table elements.

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive

exploration of customer demographics.

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
df = pd.DataFrame({  
    "Customer ID": [1, 2, 3],  
    "Age": [28, 35, 42],  
    "Gender": ["Female", "Male", "Female"],  
    "Income": [50000, 60000, 75000]  
})
```

```
print("\nCustomer Demographics")
```

```
print(df)
```

```
# 1. Bar Chart - Ages
```

```
plt.figure(figsize=(6,4))
```

```
plt.bar(df["Customer ID"].astype(str), df["Age"])
```

```
plt.title("Customer Age Distribution")
```

```
plt.xlabel("Customer ID")
```

```
plt.ylabel("Age")
```

```
plt.show()
```

```
# 2. Pie Chart - Gender
```

```
gender = df["Gender"].value_counts()
```

```
plt.figure(figsize=(5,5))
```

```
plt.pie(gender,
```

```
    labels=gender.index,
```

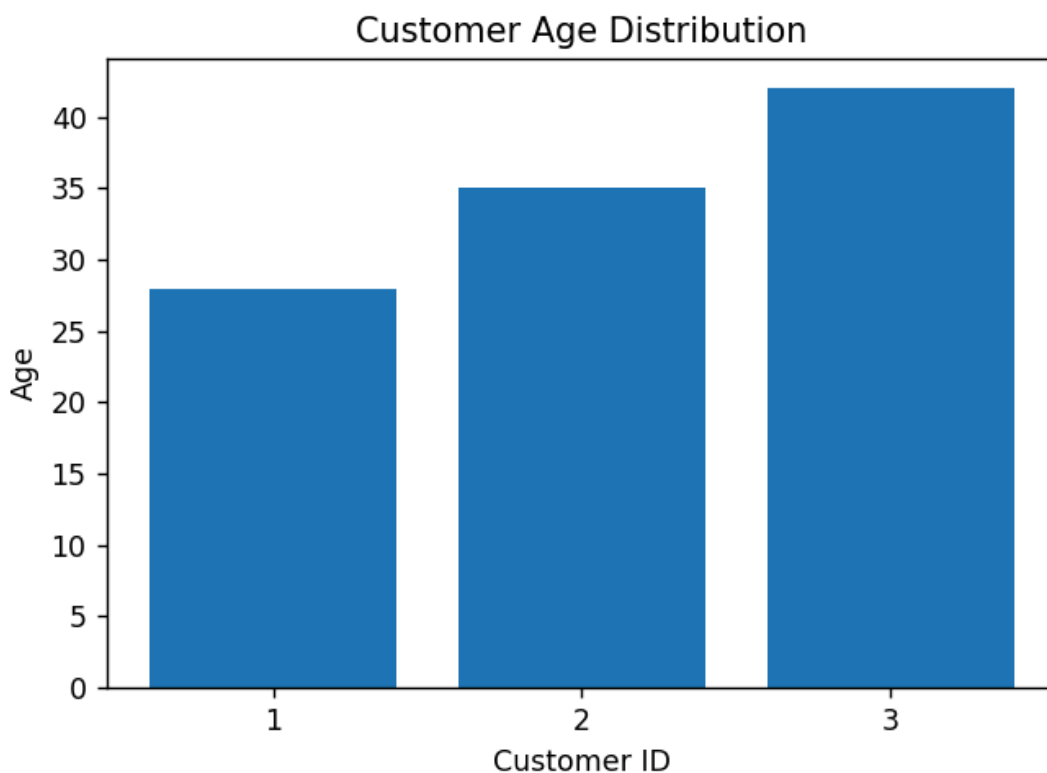
```

        autopct="%1.1f%%",
        startangle=90)
plt.title("Gender Distribution")
plt.show()

# 3. Table
fig, ax = plt.subplots(figsize=(7,2))
ax.axis("off")
table = ax.table(cellText=df.values,
                 colLabels=df.columns,
                 loc="center")
table.scale(1,2)
plt.title("Customer Demographics Table")
plt.show()

```

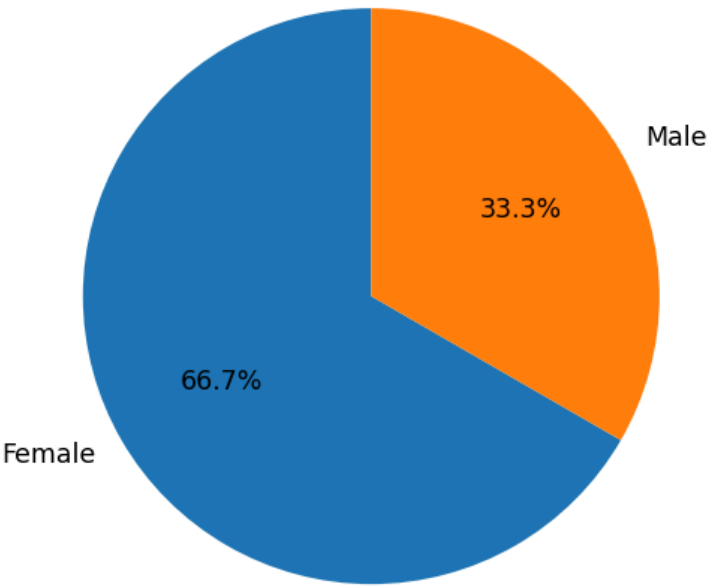
BAR CHART - CUSTOMER AGE DISTRIBUTION:



CUSTOMER DEMOGRAPHICS:

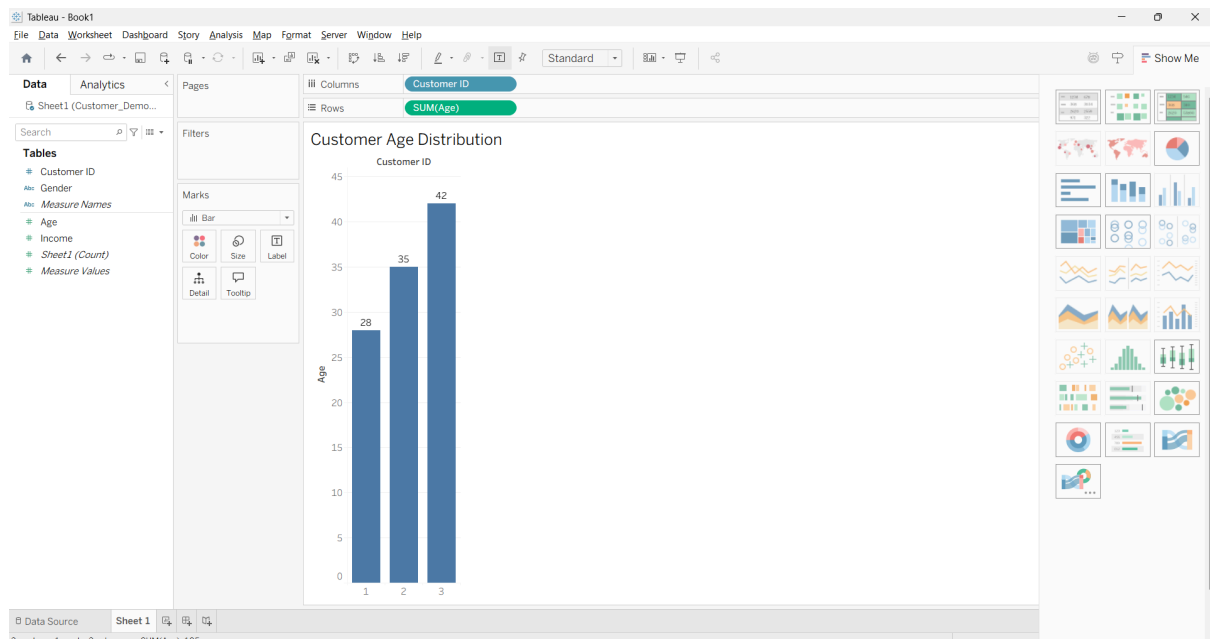
Customer Demographics				
	Customer ID	Age	Gender	Income
0	1	28	Female	50000
1	2	35	Male	60000
2	3	42	Female	75000

Gender Distribution

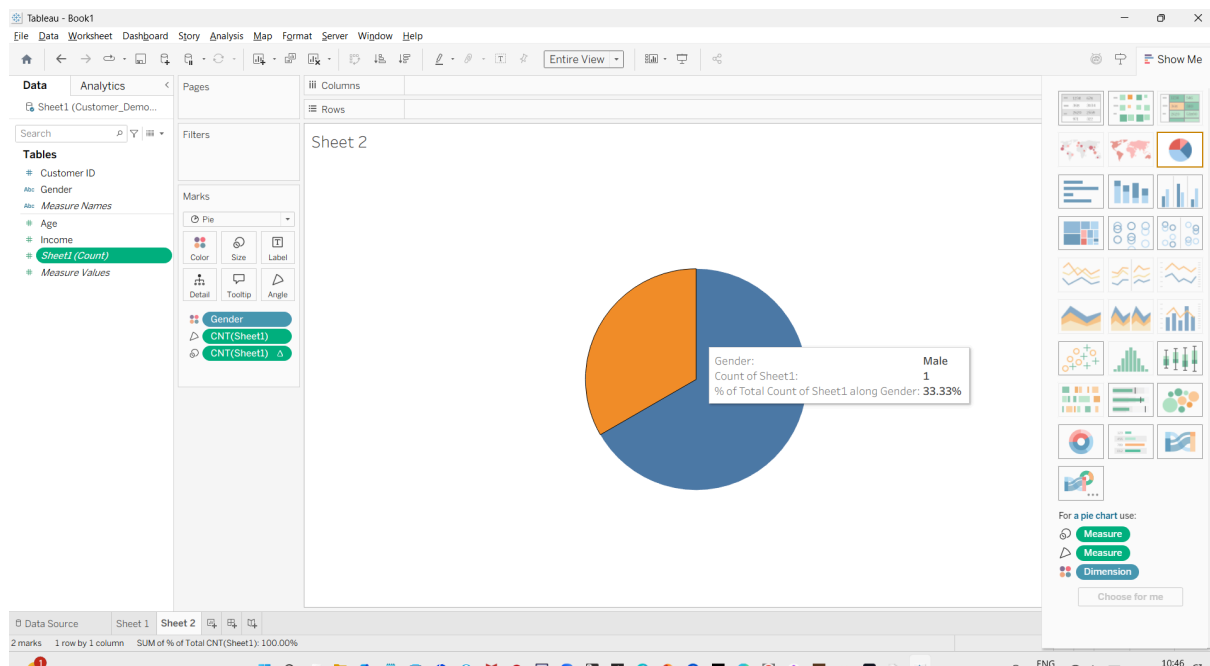


Customer Demographics Table

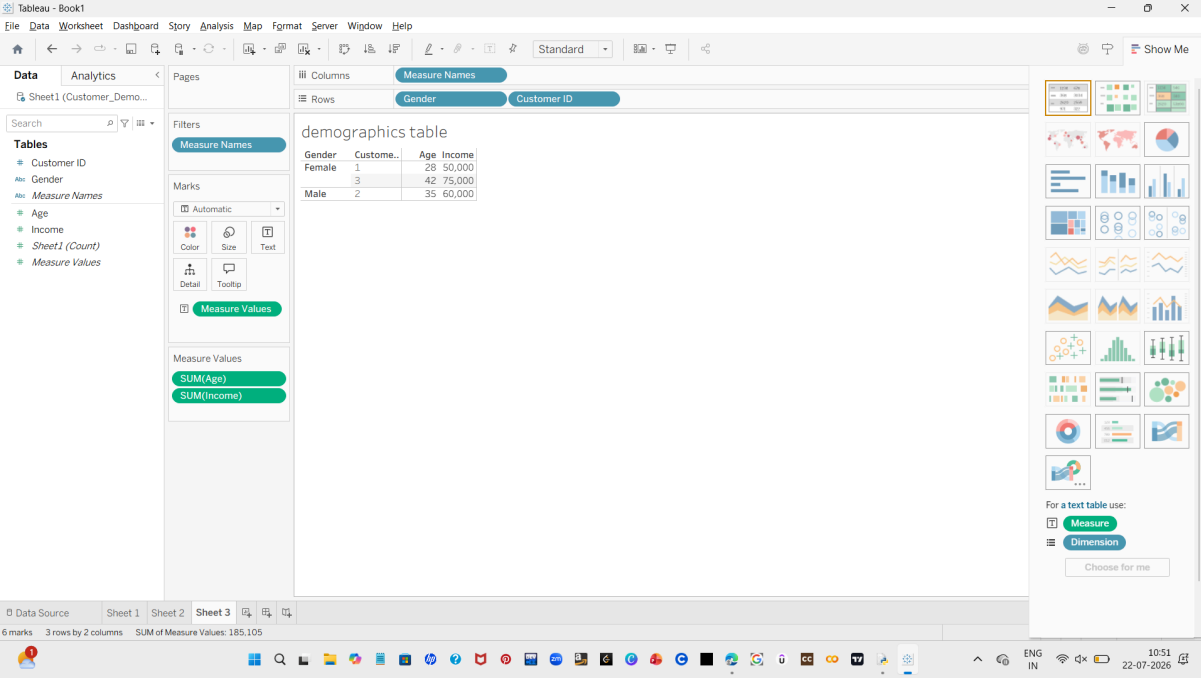
Customer ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000



Worksheet 2 – Pie Chart (Gender Distribution) - tableau



Worksheet 3 – Customer Demographics Table - Tableau



Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

3 Marketing 7 78

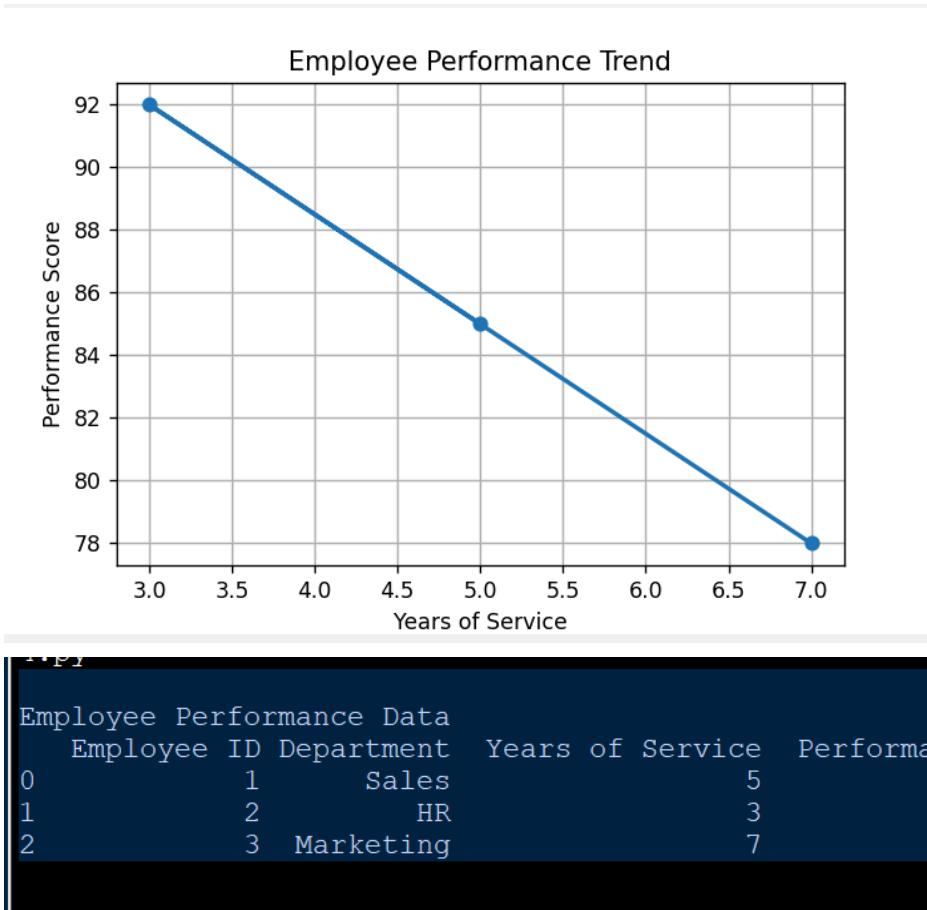
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

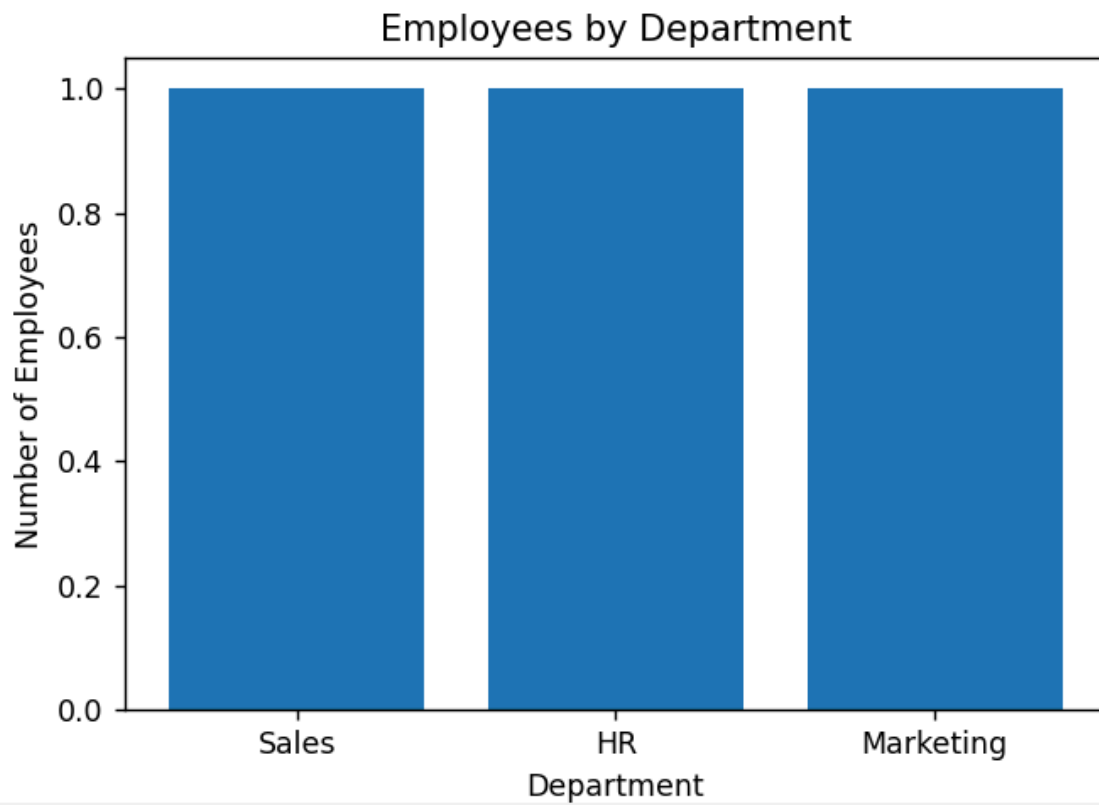
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

3. Build a table to display the performance data for each employee. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

CODE:

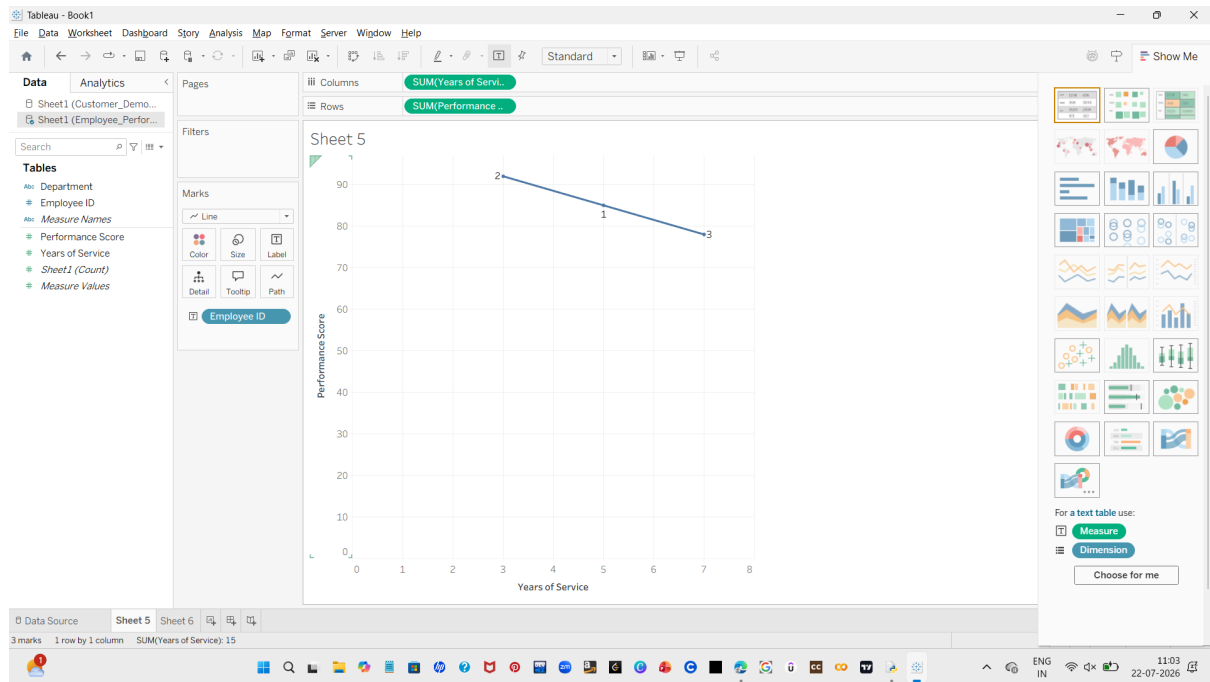




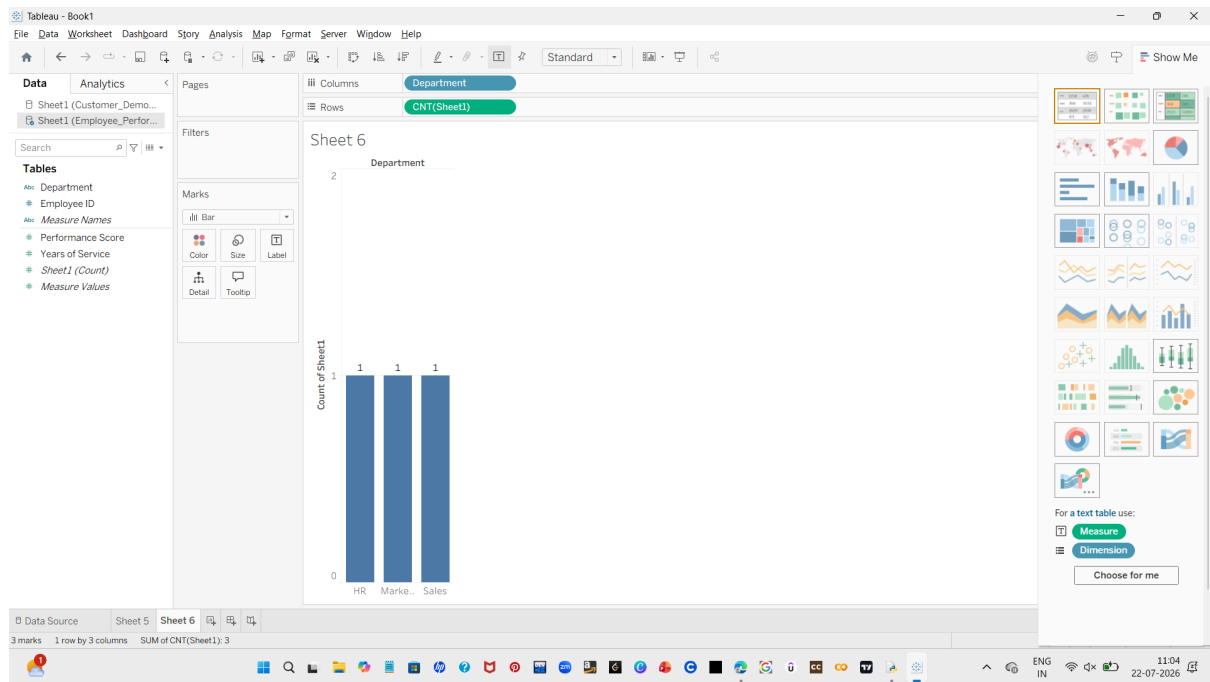
Employee Performance Table

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

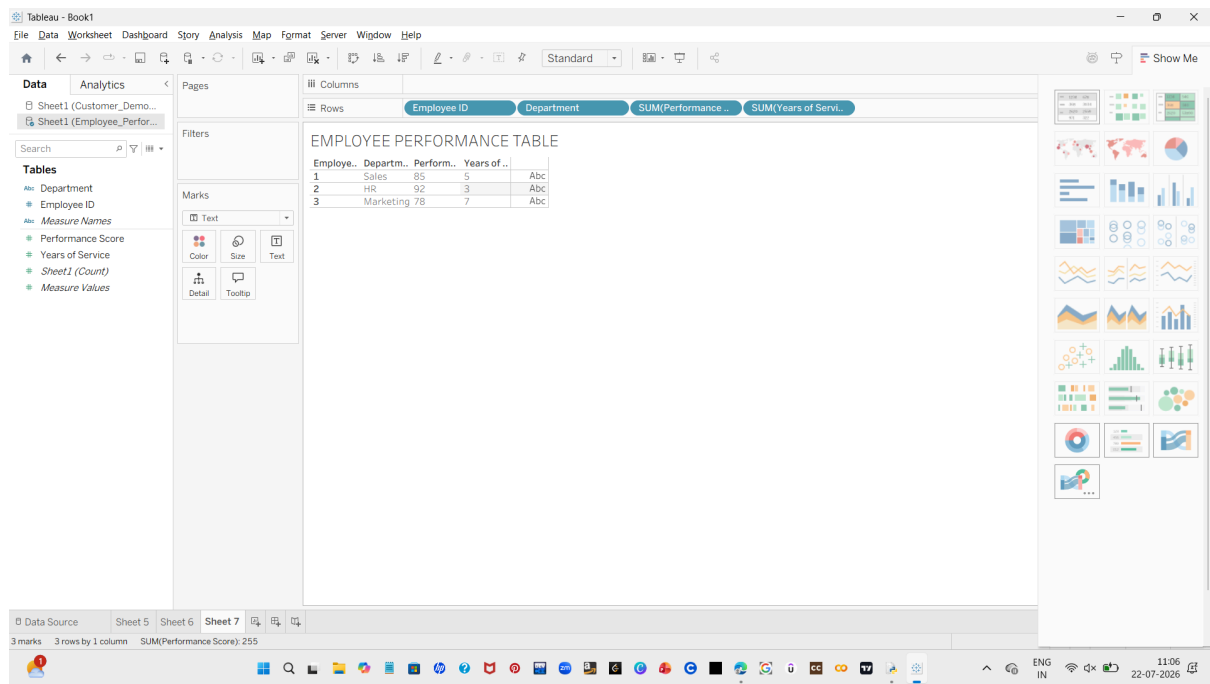
Worksheet 1: Line Chart -TABLEAU



Worksheet 2: Department Bar Chart -TABLEAU



Worksheet 3: Employee Table -TABLEAU



Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID Product Name Quantity Available Price (in \$)

1 Product A 250 20

2 Product B 175 15

3 Product C 300 18

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
df = pd.DataFrame({
    "Product ID": [1, 2, 3],
    "Product Name": ["Product A", "Product B", "Product C"],
    "Category": ["Category 1", "Category 2", "Category 1"],
    "Quantity Available": [250, 175, 300],
    "Price ($)": [20, 15, 18]
})
```

```

print(df)

# 1. Bar Chart
plt.figure(figsize=(6,4))
plt.bar(df["Product Name"], df["Quantity Available"])
plt.title("Inventory Quantity")
plt.xlabel("Product")
plt.ylabel("Quantity Available")
plt.show()

# 2. Stacked Bar Chart
pivot=df.pivot_table(index="Category",
                      columns="Product Name",
                      values="Quantity Available",
                      fill_value=0)

pivot.plot(kind="bar",stacked=True)
plt.title("Quantity by Category")
plt.xlabel("Category")
plt.ylabel("Quantity")
plt.show()

# 3. Table
fig,ax=plt.subplots(figsize=(8,2))
ax.axis("off")
table=ax.table(cellText=df.values,
               colLabels=df.columns,
               loc="center")
table.scale(1,2)
plt.title("Inventory Table")
plt.show()

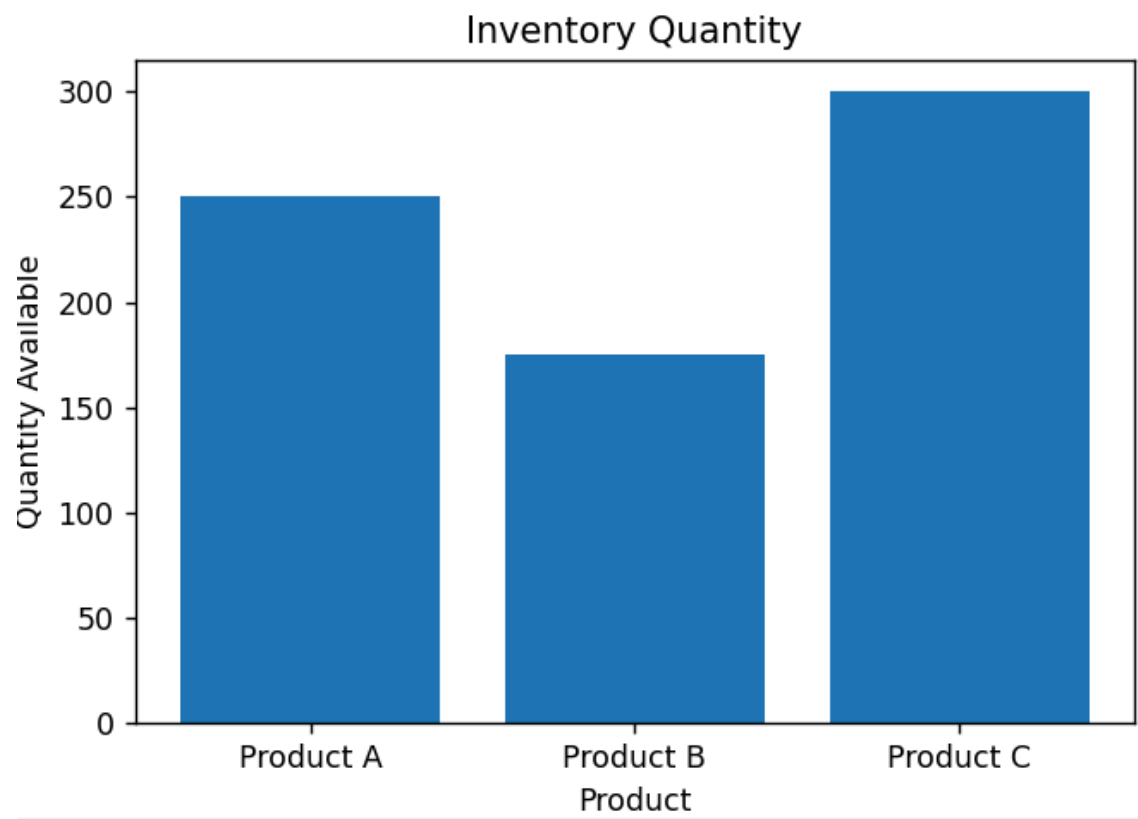
```

OUTPUT:

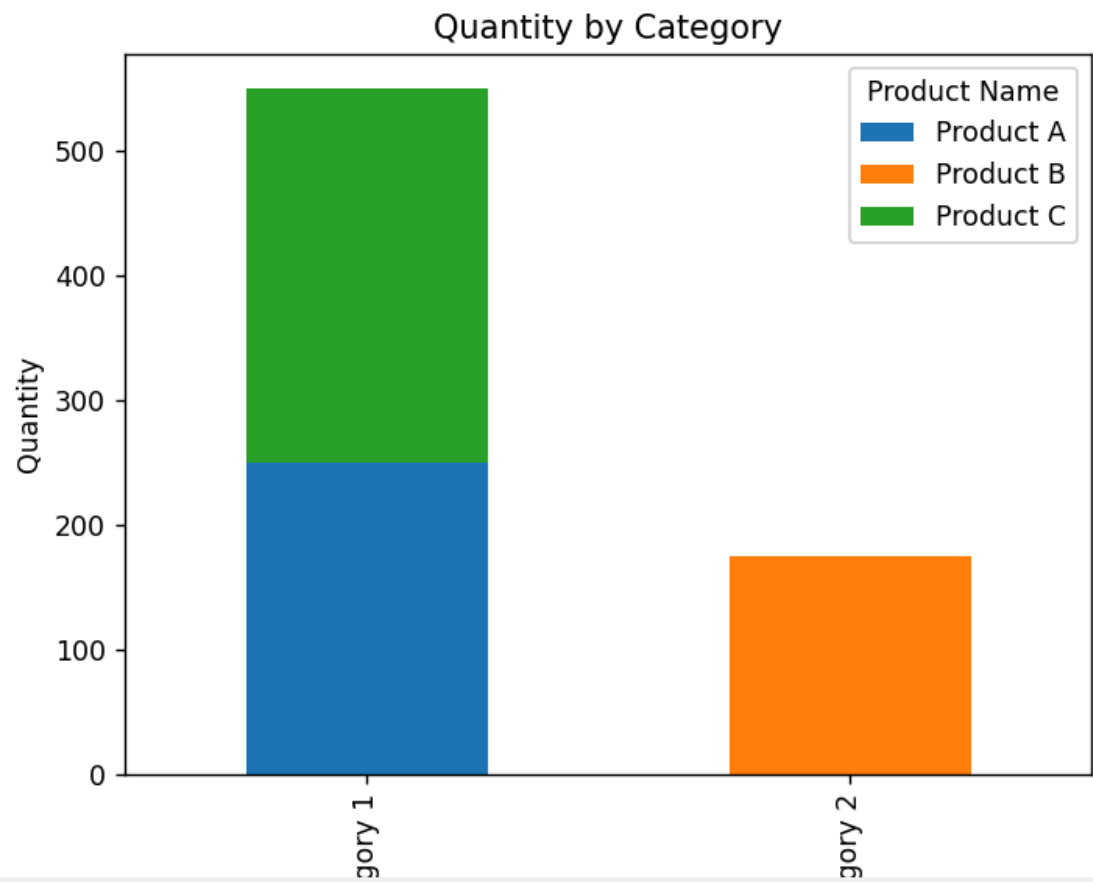
DEMOGRAPHIC TABLE

4.py	Product ID	Product Name	Category	Quantity Available	Price (\$)
0	1	Product A	Category 1	250	20
1	2	Product B	Category 2	175	15
2	3	Product C	Category 1	300	18

BAR CHART



STACKED BAR CHART



INVENTORY TABLE

Inventory table

Product ID	Product Name	Category	Quantity Available	Price (\$)
1	Product A	Category 1	250	20
2	Product B	Category 2	175	15
3	Product C	Category 1	300	18

Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

3. Build a table to show the survey response data for each survey. Label the table elements.

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table

for interactive exploration of survey responses.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df=pd.DataFrame({  
    "Survey ID":[1,2,3],  
    "Question 1":["A","B","C"],  
    "Question 2":["B","A","A"],  
    "Question 3":["C","D","B"]  
})
```

```
print(df)
```

```
# 1. Grouped Bar Chart
```

```
q1=df["Question 1"].value_counts().sort_index()
```

```
plt.figure(figsize=(6,4))  
plt.bar(q1.index,q1.values)  
plt.title("Question 1 Responses")  
plt.xlabel("Response")  
plt.ylabel("Count")  
plt.show()
```

2. Stacked Bar Chart

```
response=pd.DataFrame({  
    "Question 1":df["Question 1"].value_counts(),  
    "Question 2":df["Question 2"].value_counts(),  
    "Question 3":df["Question 3"].value_counts()  
}).fillna(0)
```

```
response.plot(kind="bar",stacked=True)
```

```
plt.title("Survey Response Distribution")  
plt.xlabel("Response")  
plt.ylabel("Count")  
plt.show()
```

3. Table

```
fig,ax=plt.subplots(figsize=(7,2))  
ax.axis("off")  
table=ax.table(cellText=df.values,  
               colLabels=df.columns,  
               loc="center")  
table.scale(1,2)  
plt.title("Survey Response Table")  
plt.show()
```

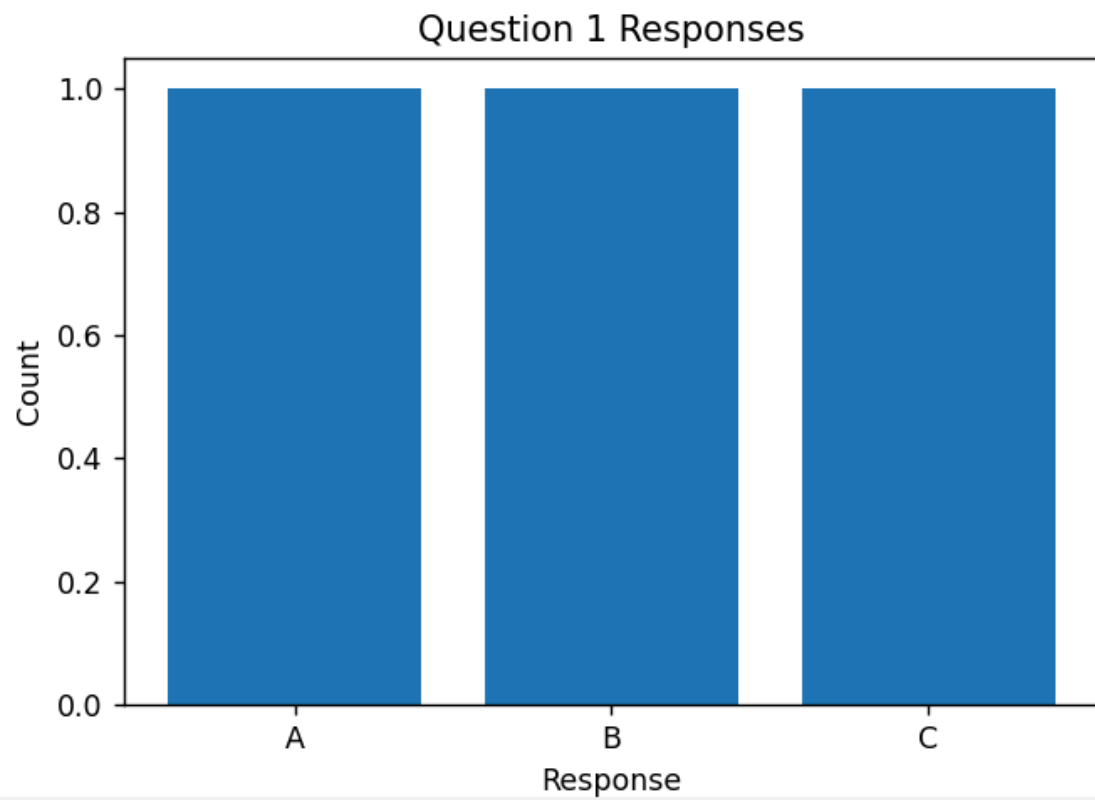
Output:

Table:

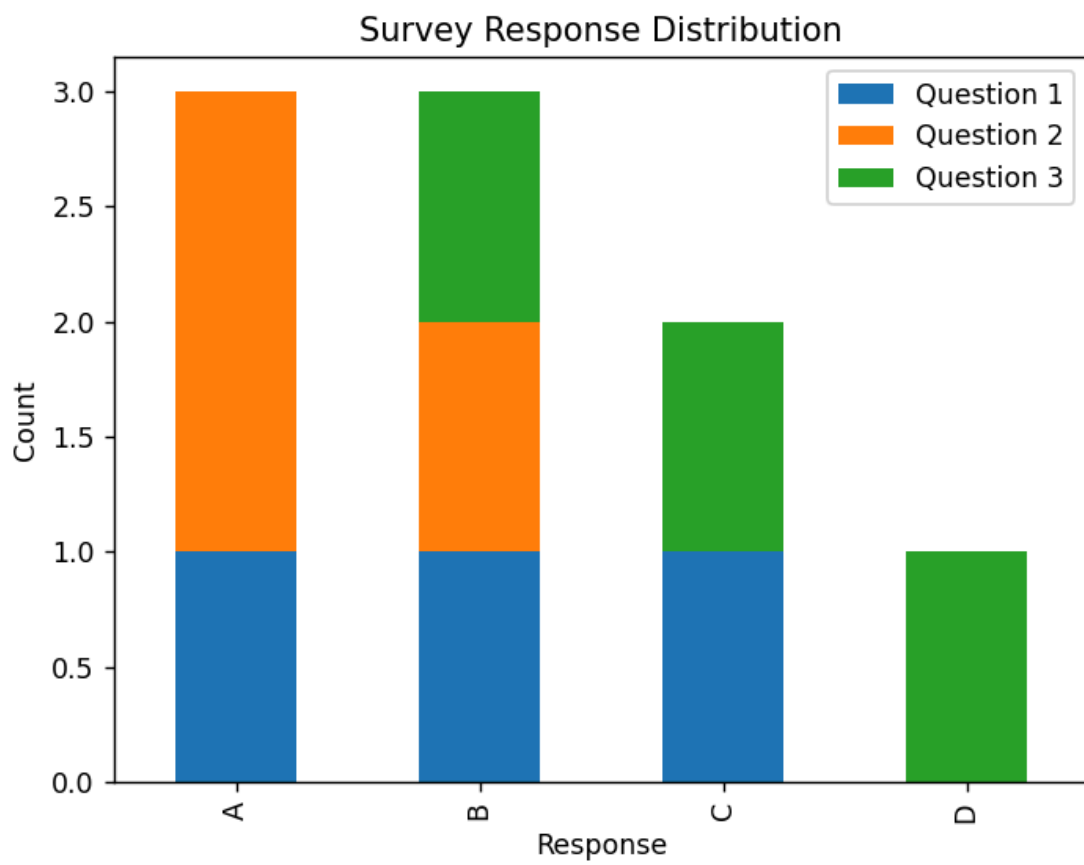
```
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4.py
```

	Survey ID	Question 1	Question 2	Question 3
0	1	A	B	C
1	2	B	A	D
2	3	C	A	B

Bar chart:



Stacked bar chart:



Survey response table

Survey Response Table

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Set 20: Stock Analysis

Dataset: Stock Prices

Date Stock A Stock B Stock C

2023-01-01 100 150 120

2023-01-02 105 152 118

2023-01-03 110 148 122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock

C) over a specific time period. Label the axes and title the chart.

2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

3. Build a table to display the stock price data for each company over the given period. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df=pd.DataFrame({  
    "Date":["2023-01-01","2023-01-02","2023-01-03"],  
    "Stock A":[100,105,110],  
    "Stock B":[150,152,148],  
    "Stock C":[120,118,122]  
})
```

```
df["Date"]=pd.to_datetime(df["Date"])
```

```
print(df)
```

```
# 1. Line Chart
```

```
plt.figure(figsize=(7,4))
```



```
plt.plot(df["Date"],df["Stock A"],marker='o',label="Stock A")
plt.plot(df["Date"],df["Stock B"],marker='o',label="Stock B")
plt.plot(df["Date"],df["Stock C"],marker='o',label="Stock C")
```

```
plt.title("Stock Prices Over Time")
plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.grid(True)
```

```
plt.show()
```

2. Daily Percentage Change

```
change=df["Stock A"].pct_change()*100
```

```
plt.figure(figsize=(6,4))
plt.bar(df["Date"].dt.strftime("%d-%m"),change.fillna(0))
```

```
plt.title("Daily Percentage Change - Stock A")
plt.xlabel("Date")
plt.ylabel("Percentage Change (%)")
```

```
plt.show()
```

3. Table

```
fig,ax=plt.subplots(figsize=(8,2))
ax.axis("off")
```

```
table=ax.table(cellText=df.values,
               colLabels=df.columns,
               loc="center")
```

```
table.scale(1,2)
```

```
plt.title("Stock Price Table")
```

```
plt.show()
```

Output:

table:

```
= RESTART: C:\Users\sclco\AppData\Local\Programs\Python\Py
4.py
```

	Date	Stock A	Stock B	Stock C
0	2023-01-01	100	150	120
1	2023-01-02	105	152	118
2	2023-01-03	110	148	122

